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Confidence Intervals/Sets

Methods of finding $(1-\alpha)$ confidence set

Method 1 : $X \sim f(x|\theta) ; \theta \in \Theta$

$Q(X, \theta)$ = Pivot for θ | if the distribution of Q does not depend on θ

1) Find a suitable pivot $Q(X, \theta)$

2) Find constants a, b

st. $P_{\theta}(a \leq Q(X, \theta) \leq b) = 1-\alpha$

3) Define for each sample point x

$$C(x) = \{ \theta \in \Theta \mid a \leq Q(x, \theta) \leq b \}$$

A random set $C(X)$ is a $(1-\alpha)$ level confidence set

Remark 1 : If $Q(x, \theta)$ is increasing or decreasing in θ , then

$C(x)$ is an interval otherwise

$C(x)$ is set not an interval.

Remark 2 : If we take $a = -\infty$ or $b = \infty$ then we get one-sided intervals

Examples: $X_1, X_2, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$

Find CI for μ

(i) Pivot $Q(x, \theta) = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}$

(ii) $a = t_{n-1, \alpha/2}$

$b = t_{n-1, 1-\alpha/2}$

(iii) $C(x) = \{ \mu \mid t_{n-1, \alpha/2} \leq \frac{\bar{x} - \mu}{s/\sqrt{n}} \leq t_{n-1, 1-\alpha/2} \}$

$= \bar{x} \pm \frac{s}{\sqrt{n}} t_{n-1, 1-\alpha/2}$

Suppose we want ^{upper} ~~one~~ confidence interval for μ

Take $b = \infty$

$P\left(a < \frac{\sqrt{n}(\bar{x} - \mu)}{s}\right) = 1 - \alpha$

$\Rightarrow a = t_{n-1, \alpha}$

$C(x) = \{ \mu \mid t_{n-1, \alpha} < \frac{\sqrt{n}(\bar{x} - \mu)}{s} \}$

$= \left(-\infty, \bar{x} - t_{n-1, \alpha} \frac{s}{\sqrt{n}} \right]$

For lower confidence interval.

$\left[\bar{x} - t_{n-1, 1-\alpha} \frac{s}{\sqrt{n}}, \infty \right)$

Find CI for σ^2

(i) Pivot $Q = \frac{(n-1)S^2}{\sigma^2}$

(ii) $a = \chi^2_{n-1, \alpha/2}$

$b = \chi^2_{n-1, 1-\alpha/2}$

(iii) $CI = \left\{ \sigma^2 \mid \chi^2_{n-1, \alpha/2} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{n-1, 1-\alpha/2} \right\}$

~~$\chi^2_{n-1, \alpha/2} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{n-1, 1-\alpha/2}$~~

$= \left[\frac{(n-1)S^2}{\chi^2_{n-1, 1-\alpha/2}}, \frac{(n-1)S^2}{\chi^2_{n-1, \alpha/2}} \right]$

Example 1: $X_1, X_2, \dots, X_n \sim \text{Unif}(0, \theta)$

(i) $Q = \frac{X_{(n)}}{\theta}$ because $P\left(\frac{X_{(n)}}{\theta} \leq x\right) = x^n$

(ii) a, b st $P\left(a \leq \frac{X_{(n)}}{\theta} \leq b\right) = 1 - \alpha$

$\Rightarrow b^n - a^n = 1 - \alpha$

Note $\left[\frac{X_{(n)}}{\theta} \sim \text{Beta}(n, 1) \right]$

$$(ii) C(x) = \left\{ \theta \mid b_{\alpha/2} \leq \frac{x_{(n)}}{\theta} \leq b_{1-\alpha/2} \right\}$$

$$= \left[\frac{x_{(n)}}{b_{1-\alpha/2}}, \frac{x_{(n)}}{b_{\alpha/2}} \right]$$

where $b_p = p$ -th quantile of $\text{Beta}(n, 1)$

Method-II : Inverting a test

For any $\theta \in \Theta_0$. Suppose $A(\theta_0)$ = acceptable region of a level- α test

for testing $H_0: \theta = \theta_0$ vs $H_1: \neg H_0$

Inverting a test $\Rightarrow$ define:

$$C(x) = \left\{ \theta_0 \in \Theta \mid x \in A(\theta_0) \right\}$$

Claim: $C(x)$ is $(1-\alpha)$ confidence set.

Example: $x_1, x_2, \dots, x_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

Find lower confidence for μ

$H_0: \mu = \mu_0$ vs $H_1: \mu > \mu_0$

$$\phi(x) = \mathbb{1} \left\{ \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} > z_{n-1, 1-\alpha} \right\}$$

$$A(\underline{\mu}_0) = \{ \underline{x} \mid \frac{(\bar{x} - \underline{\mu}_0) \sqrt{n}}{s} < t_{n-1, 1-\alpha} \}$$

$$C(\underline{x}) = \{ \underline{\mu}_0 \mid \frac{\sqrt{n} (\bar{x} - \underline{\mu}_0)}{s} < t_{n-1, 1-\alpha} \}$$

$$= \left[\bar{x} - \frac{s}{\sqrt{n}} t_{n-1, 1-\alpha}, \infty \right)$$

Theorem: [Relation between test - confidence set.]

$$X \sim f(x|\theta) ; \theta \in \Theta \subseteq \mathbb{R}^k$$

(i) [test $\rightarrow$ confidence set] Suppose $\forall \theta_0 \in \Theta$

$A(\theta_0)$ is the acceptance region of a level- α test for $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$

~~The random set $C(\underline{x})$ is a $(1-\alpha)$ confidence set~~

For each sample point $\underline{x}$ define:

$$C(\underline{x}) = \{ \theta_0 \in \Theta \mid \underline{x} \in A(\theta_0) \}$$

The random set $C(\underline{x})$ is a $(1-\alpha)$ confidence set
 (that is, $\inf_{\theta \in \Theta} P_{\theta}(\theta \in C(\underline{x})) \geq 1-\alpha$)

(ii) [confidence set $\rightarrow$ test] Suppose $C(\underline{x})$ is a $(1-\alpha)$ confidence set for θ . For each $\theta_0 \in \Theta$ define $A(\theta_0) = \{ \underline{x} \mid \theta_0 \in C(\underline{x}) \}$. Then $A(\theta_0)$

Then $A(\theta_0)$ is an acceptance region for a level- α test for testing $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$.

Proof: (i) Fix $\theta_0 \in \Theta$

$$\begin{aligned} P_{\theta_0}(\theta_0 \in C(x)) &= P_{\theta_0}(x \in A(\theta_0)) \\ &= 1 - P_{\theta_0}(x \in R(\theta_0)) \\ &\leq \alpha. \end{aligned}$$

$$\Rightarrow \inf_{\theta_0 \in \Theta} P_{\theta_0}(\theta_0 \in C(x)) \geq 1 - \alpha$$

$$\begin{aligned} \text{(ii)} \quad P_{\theta_0}(A \in C(\theta_0)) &= P_{\theta_0}(\theta_0 \in C(x)) \geq 1 - \alpha \quad \forall \theta_0 \\ \Rightarrow P_{\theta_0}(x \in R(\theta_0)) &\leq \alpha \end{aligned}$$

Example: $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$

Test $H_0: \mu = 1$ vs $H_1: \mu \neq 1$

Suppose ~~the~~ the observe $(1-\alpha)$ CI for μ is $[-1, 1.5]$ $\Leftrightarrow x \in A(\mu_0)$

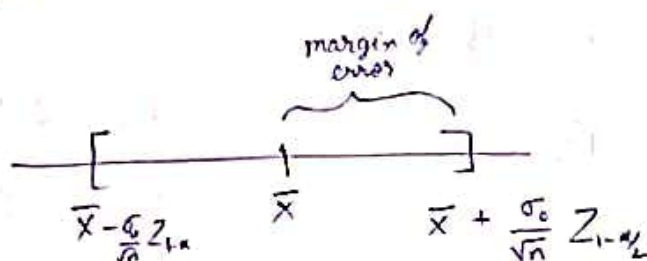
Accept H_0

For the test $H_0: \mu = 2$ vs $H_1: \mu \neq 2$

Reject H_0

Example $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma_0^2)$
 $\hookrightarrow$ known

Confidence interval for $\bar{X} \pm \frac{\sigma_0}{\sqrt{n}} Z_{1-\frac{\alpha}{2}}$



Note: (i) Precision of a confidence set is given by the ~~ex~~ volume of the set.

(ii) Coverage ~~prob~~ probability $\uparrow$

$\Leftrightarrow$ width of interval $\uparrow$ (fixed n)

$\Leftrightarrow$ sample size (n) $\uparrow$ (fixed width)

An optimality criteria:

Among all $(1-\alpha)$ level confidence intervals choose the one that has the "shortest length"

Example: $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma_0^2)$
 $\hookrightarrow$ known

$$Q(\bar{x}, \mu) = \frac{\bar{x} - \mu}{\sigma_0 / \sqrt{n}}$$

$$P\left(a \leq \frac{\bar{X} - \mu}{\sigma_0 / \sqrt{n}} \leq b\right) = 1 - \alpha$$

$$\Rightarrow P\left(\bar{X} - b \frac{\sigma_0}{\sqrt{n}} \leq \mu \leq \bar{X} + a \frac{\sigma_0}{\sqrt{n}}\right) = 1 - \alpha$$

Confidence interval $\left[\bar{X} - b \frac{\sigma_0}{\sqrt{n}}, \bar{X} - \frac{a \sigma_0}{\sqrt{n}} \right]$

$W = \frac{\sigma_0}{\sqrt{n}} (b-a) \rightarrow \text{width}$

minimizing $W \Leftrightarrow$ minimising $(b-a)$

Example: $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

$Q = \frac{(n-1)S^2}{\sigma^2}$

$P\left(a \leq \frac{(n-1)S^2}{\sigma^2} \leq b\right) = 1-\alpha$

$\Rightarrow P\left(\frac{(n-1)S^2}{b} \leq \sigma^2 \leq \frac{(n-1)S^2}{a}\right) = 1-\alpha$

Confidence interval $\left[\frac{(n-1)S^2}{b}, \frac{(n-1)S^2}{a} \right]$

$W = (n-1)S^2 \left(\frac{1}{a} - \frac{1}{b} \right) \rightarrow \text{width}$

minimizing $W \Leftrightarrow$ minimising $\left(\frac{1}{a} - \frac{1}{b} \right)$

Theorem: (i) Suppose $X \sim f(x|\theta)$; $\theta \in \Theta \subseteq \mathbb{R}^k$.

Suppose for a +ve statistic $S(x)$ and real valued statistic $T(x)$ and $Q = \frac{T-\theta}{S}$

is a pivotal quantity for θ .

Suppose Q has a pdf f that is unimodal at

$$\left[f \text{ is non-dec } \forall x \leq x_0, \text{ } f \text{ is non-inc } \forall x \geq x_0 \right]$$

Consider the class

$$G = \{ [T-bS, T-aS] \mid a, b \in \mathbb{R}, \int_a^b f(x) dx = 1 \}$$

If $a^*, b^* \in \mathbb{R}$ and are st.

$$(i) [T-b^*S, T-a^*S] \in G$$

$$(ii) f(a^*) = f(b^*) > 0$$

$$(iii) a^* \leq x_0 \leq b^*$$

Then $[T-b^*S, T-a^*S]$ has the shortest length within class G

Suppose $T(x) > 0$; $\theta > 0$ and $Q = T/\theta$ is a pivot for θ . Suppose f is the pdf of Q is such that $x^2 f(x)$ is unimodal at x_0

Consider the class

$$G = \left\{ \left[\frac{T}{b}, \frac{T}{a} \right] \mid a, b > 0; \int_a^b f(x) dx = 1 \right\}$$

If $a^*, b^* \in \mathbb{R}$ and are st.

$$(i) \left[\frac{T}{b^*}, \frac{T}{a^*} \right] \in G$$

$$(ii) (a^*)^2 f(a^*) = (b^*)^2 f(b^*) > 0$$

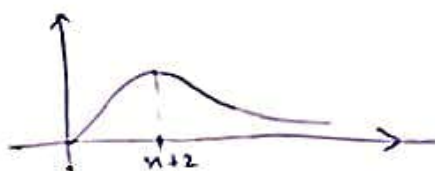
$$(iii) \quad a^* \leq x_0 \leq b^*$$

Then $\left[\frac{T}{b^*}, \frac{T}{a^*}\right]$ has the shortest length within class C

Example: $X_1, X_2, \dots, X_n \sim \mathcal{N}(0, \sigma^2)$

$$Q = \frac{T}{\sigma^2} = \frac{\sum X_i^2}{\sigma^2} \sim \chi_n^2$$

$$\therefore x^2 f(x) = \frac{e^{-\frac{x}{2}} x^{\frac{n}{2}+1}}{\Gamma(\frac{n}{2}) 2^{n/2}}$$



unimodal
at $\underline{n+2}$

$\therefore \left[\frac{\sum X_i^2}{b^*}, \frac{\sum X_i^2}{a^*}\right]$ is the shortest CI for σ^2

where a^*, b^* satisfy

$$(i) \quad \int_{a^*}^{b^*} f(x) dx = 1 - \alpha$$

$$(ii) \quad (a^*)^2 f(a^*) = (b^*)^2 f(b^*)$$

Not Solvable

Example: $X_1, X_2, \dots, X_{n_1} \stackrel{iid}{\sim} \mathcal{N}(\mu_1, \sigma^2)$
 $Y_1, Y_2, \dots, Y_{n_2} \stackrel{iid}{\sim} \mathcal{N}(\mu_2, \sigma^2)$ $\rightarrow$ independent

Stats CI of $(\mu_1 - \mu_2)$

$$Q = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where $S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$

$Q \sim t_{n_1 + n_2 - 2}$

Based on Q the interval for $(\mu_1 - \mu_2)$

$$\left[(\bar{X} - \bar{Y}) - b S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, (\bar{X} - \bar{Y}) - a S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right]$$

$$\text{length} = (b - a) S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Find a^*, b^* (i) $\int_{a^*}^{b^*} f(u) du = 1 - \alpha$

(ii) $f(a^*) = f(b^*) > 0$

$$\begin{cases} a^* = t_{n_1 + n_2 - 2, \alpha/2} \\ b^* = t_{n_1 + n_2 - 2, 1 - \alpha/2} \end{cases}$$

Note:
(idea)

$$\frac{d}{da} \left(\int_a^b f(u) du \right) = 0 \Rightarrow \frac{db}{da} = \frac{f(a)}{f(b)}$$

$$\frac{d}{db} \left(\frac{1}{a} - \frac{1}{b} \right) = 0 \Rightarrow \frac{db}{da} = \frac{b^2}{a^2}$$

$$\therefore \frac{f(a)}{f(b)} = \frac{b^2}{a^2} \Rightarrow a^2 f(a) = b^2 f(b)$$

Problem: Cannot find shortest length CI for one-sided intervals.

Test Related Optimality

$$X \sim f(x|\theta) ; \theta \in \Theta \subseteq \mathbb{R}^k$$

We know that true coverage probability of a confidence set $C(x)$ is $= P_\theta(\theta \in C(x))$

~~$P_{\theta'}(\theta \in C(x))$~~

The probability of false coverage $= P_{\theta'}(\theta' \in C(x))$

for $\theta' \neq \theta$

Theorem: Let $X \sim f(x|\theta) ; \theta \in \Theta \subseteq \mathbb{R}^k$ and $C(x)$ is a confidence set for θ

$$\text{if } \text{Vol}(C(x)) < \infty \text{ w.p. 1}$$

$$E_\theta [\text{Vol}(C(x))] = \int_\Theta P_{\theta'}(\theta' \in C(x)) d\theta'$$

Proof:
$$\int_{\mathcal{X}} \text{vol}(C(x)) f(n|0) dx$$

Subst:
$$\int_{\Theta} \int_{\mathcal{X}} \mathbb{1}(0' \in C(x)) f(n|0) dx d0'$$

$$= \int_{\Theta} P_0(0' \in C(x)) d0'$$

Notes: Shorter interval tend to cover less no. of false parameter values. So we minimize the probability of false coverage

Def: A confidence set $C^*(x)$ of θ is called Uniformly most accurate (UMA) ~~(UMC)~~ $(1-\alpha)$ confidence set if it minimizes

[uniformly] ~~the~~ the probability of false coverage among all $(1-\alpha)$ level confidence sets.

(i) $P_0(\theta \in C^*(x)) \geq 1-\alpha$

(ii) $P_0(0' \in C^*(x)) \leq P_0(0' \in C(x))$

$\forall 0' \neq \theta$ (defined earlier)
and any $(1-\alpha)$ level
Confidence Set $C(x) \quad \forall \theta \in \Theta$

Theorem: Suppose $\forall \theta_0 \in \Theta$, $A^*(\theta_0)$ = acceptance region of a UMP level- α test

for testing $H_0: \theta = \theta_0 \quad \text{vs} \quad H_1: \theta \neq \theta_0$

The set $C^*(x) = \{\theta_0 \in \Theta_0 \mid x \in A^*(x)\}$

is a ~~level (1- α) UMP~~ level $(1-\alpha)$ UMA confidence set for θ

Proof: $C(x) \Rightarrow (1-\alpha)$ level confidence set

Define $A(\theta_0) = \{x \mid \theta_0 \in C(x)\}$

$$P_{\theta_0}(x \in A(\theta_0)) = P_{\theta_0}(\theta_0 \in C(x)) \geq 1-\alpha$$

For any $\theta \neq \theta_0$

$$P_{\theta}(\theta_0 \in C^*(x)) = P_{\theta}(x \in A^*(\theta_0))$$

$$= 1 - P_{\theta}(x \in R^*(\theta_0))$$

$$\text{power condition} \cdot \left\{ \begin{array}{l} \leq 1 - P_{\theta}(x \in R(\theta_0)) \\ = P_{\theta}(x \in A(\theta_0)) \\ = P_{\theta}(\theta_0 \in C(x)) \end{array} \right.$$

$$= P_{\theta}(x \in A(\theta_0))$$

$$= P_{\theta}(\theta_0 \in C(x))$$

Def: Let $X \sim f(x|\theta)$; $\theta \in \Theta \subseteq \mathbb{R}^k$
and $\Theta' \subseteq \Theta$ such that Θ' does not
contain the true parameter θ .

A $(1-\alpha)$ level confidence set $C(x)$
is called Θ' -unbiased if

false coverage probability

$$= P_{\theta}(\theta' \in C(x)) \leq 1 - \alpha$$

$$\forall \theta' \in \Theta'$$

$C(x)$ is called unbiased if ~~Θ'~~

$$\Theta' = \{\theta' | \theta' \neq \theta\}$$

Def: A confidence set $C^*(x)$ is uniformly most accurate unbiased (UMAU) θ

$(1-\alpha)$ confidence set for θ if

$$(i) \inf_{\theta \in \Theta} P_{\theta}(\theta \in C^*(x)) \geq 1 - \alpha$$

$$(ii) P_{\theta}(\theta' \in C^*(x)) \leq 1 - \alpha \quad \forall \theta' \neq \theta$$

$$(iii) P_{\theta}(\theta' \in C^*(x)) \leq P_{\theta}(\theta' \in C(x))$$

$$\forall \theta' \neq \theta$$

and any (a.s.)
unbiased confidence set

